

NOTE: PRACTICAL INTERPOLATION:

(i). Interpolation for beginning or end of a table, Newton's forward and backward formulae are useful, because higher order central differences do not exist at the beginning or end of a table of values.

(ii) For interpolation near the middle of a table:

(a) Stirling formula gives most accurate result for $-\frac{1}{4} \leq p \leq \frac{1}{4}$

(b) Bessel's formula most efficient ~~for~~ near $p = \frac{1}{2}$, say $\frac{1}{4} \leq p \leq \frac{3}{4}$.

(iii) If the highest ~~order~~ difference is of odd order, Stirling formula is recommended, if it is even order, Bessel's formula might be preferred.

(iv) In Stirling formula, the term containing the third differences, viz, $\frac{p(p^2-1)}{6} \cdot \frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2}$ may be neglected if $\left| \frac{p(p^2-1)}{6} \cdot \frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right| < 1$

for all p in the range $0 \leq p \leq 1$.

Maximum value of $p \frac{(p^2-1)}{6}$ is 0.064 , so

$$|0.064 \cdot \frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2}| < \frac{1}{2} \Rightarrow \left| \frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right| < 8.$$

For Bessel's, $\left| \frac{p(p-1)(p-\frac{1}{2})}{6} \Delta^3 y_{-1} \right| < \frac{1}{2}$

$$\Rightarrow \left| \Delta^3 y_{-1} \right| < 60 \quad \left[\text{since Max}^m \text{ value of } p(p-1)(p-\frac{1}{2}) \text{ is } 0.008 \right]$$

(*) If we neglect the third differences, Bessel's formula is about 7 times more accurate than Stirling's formula.

BESSEL'S INTERPOLATION FORMULA:

$$y(x) = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \cdot \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \frac{p(p-1)(p-\frac{1}{2})}{3!} \Delta^3 y_{-1} \\ + \frac{(p+1)p(p-1)(p-2)}{4!} \frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \dots$$

Example: The following table gives the values of e^x for certain equidistant values of x . Find the value of e^x when $x=0.644$

x	0.61	0.62	0.63	0.64	0.65	0.66	0.67
$y=e^x$	1.84 1.840431	1.858928	1.877610	1.896481	1.915541	1.934792	1.954237

Solution: The table of difference is

x	y	Δ	Δ^2	Δ^3	Δ^4
(x_{-3}) 0.61	(y_{-3}) 1.840431	$\Delta y_{-3} = 0.018497$	$\Delta^2 y_{-3} = 0.000185$	$\Delta^3 y_{-3} = 0.000004$	
(x_{-2}) 0.62	(y_{-2}) 1.858928	$\Delta y_{-2} = 0.018682$	$\Delta^2 y_{-2} = 0.000189$	$\Delta^3 y_{-2} = 0$	$\Delta^4 y_{-3} = -0.000004$
(x_{-1}) 0.63	(y_{-1}) 1.877610	$\Delta y_{-1} = 0.018871$	$\Delta^2 y_{-1} = 0.000189$	$\Delta^3 y_{-1} = 0.000002$	$\Delta^4 y_{-2} = 0.000002$
(x_0) 0.64	(y_0) 1.896481	$\Delta y_0 = 0.019060$	$\Delta^2 y_0 = 0.000191$	$\Delta^3 y_0 = 0.000003$	$\Delta^4 y_{-1} = 0.000001$
(x_1) 0.65	(y_1) 1.915541	$\Delta y_1 = 0.019251$	$\Delta^2 y_1 = 0.000194$		
(x_2) 0.66	(y_2) 1.934792	$\Delta y_2 = 0.019445$			
(x_3) 0.67	(y_3) 1.954237				

Here $p = \frac{0.644 - 0.64}{0.01} = 0.4$

The third difference contribution to both Stirling's and Bessel's formulae is negligible, we obtain using Stirling's formula:

$$y(0.644) = 1.896481 + 0.4 \frac{0.018871 + 0.019060}{2} + \frac{0.16}{2} (0.000189) = 1.904082$$

Using Bessel's formula:

$$y(0.644) = 1.896481 + 0.4 (0.019060) + \frac{0.4(0.4-1)}{2} \left(\frac{0.000189 + 0.000191}{2} \right) = 1.904082$$